BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

ASSIGNMENT DOCUMENTATION
SOFTWARE SYSTEM SPECIFICATION & ARCHITECTURAL MODELING

Course Name & Code	SS ZG562: Software Engineering and Management
Project / System Title	Smart Construction Site Material & Inventory Management System
Group Name & Members	**SiteSathi** | Ankit Kushwaha, Manoj Gadamsetty
Document Scope	Complete Software Requirements Specification, Use Case Analysis, Architectural Design, Domain Class Modeling, Behavioral UML Diagrams

1. Executive Summary & Assignment Scope

Civil construction projects frequently encounter significant operational bottlenecks, cost overruns, and project delays due to manual, paper-based tracking of essential raw materials such as cement, structural steel, ready-mix concrete, and aggregates. In multi-site civil engineering projects, lack of real-time inventory visibility leads to material theft, stockout-induced labor idle time, over-procurement, and wastage. To address these industry-wide challenges, this document presents the complete software engineering requirements specification and architectural design for the **Smart Construction Site Material & Inventory Management System**.

Material Wastage
-20%
Reduction in raw site wastage & theft within 6 months

Stockout Idle Time
-35%
Decrease in work stoppages via auto-reorders

PO Lead Time
< 4 Hours
Turnaround time from indent to supplier dispatch

Audit Accuracy
98%
Match between physical counts & digital log

SYSTEM MOTIVE & CORE OPERATIONAL PILLARS AT A GLANCE

• Operational Problem: Civil sites lose 15-20% raw materials (cement, steel, aggregates) due to paper tracking, theft, and stockout delays.
• Solution Architecture: A 3-Tier Layered Application (React Native PWA + Node.js REST API + PostgreSQL) featuring offline mobile synchronization for zero-connectivity field yards.
• Core Value Driver: Automates safety stock monitoring, QR gate docket verification, and instant reorder alerts (<<extend>> Low-Stock Alert).

1.1 High-Level System Motive & Use Case Map

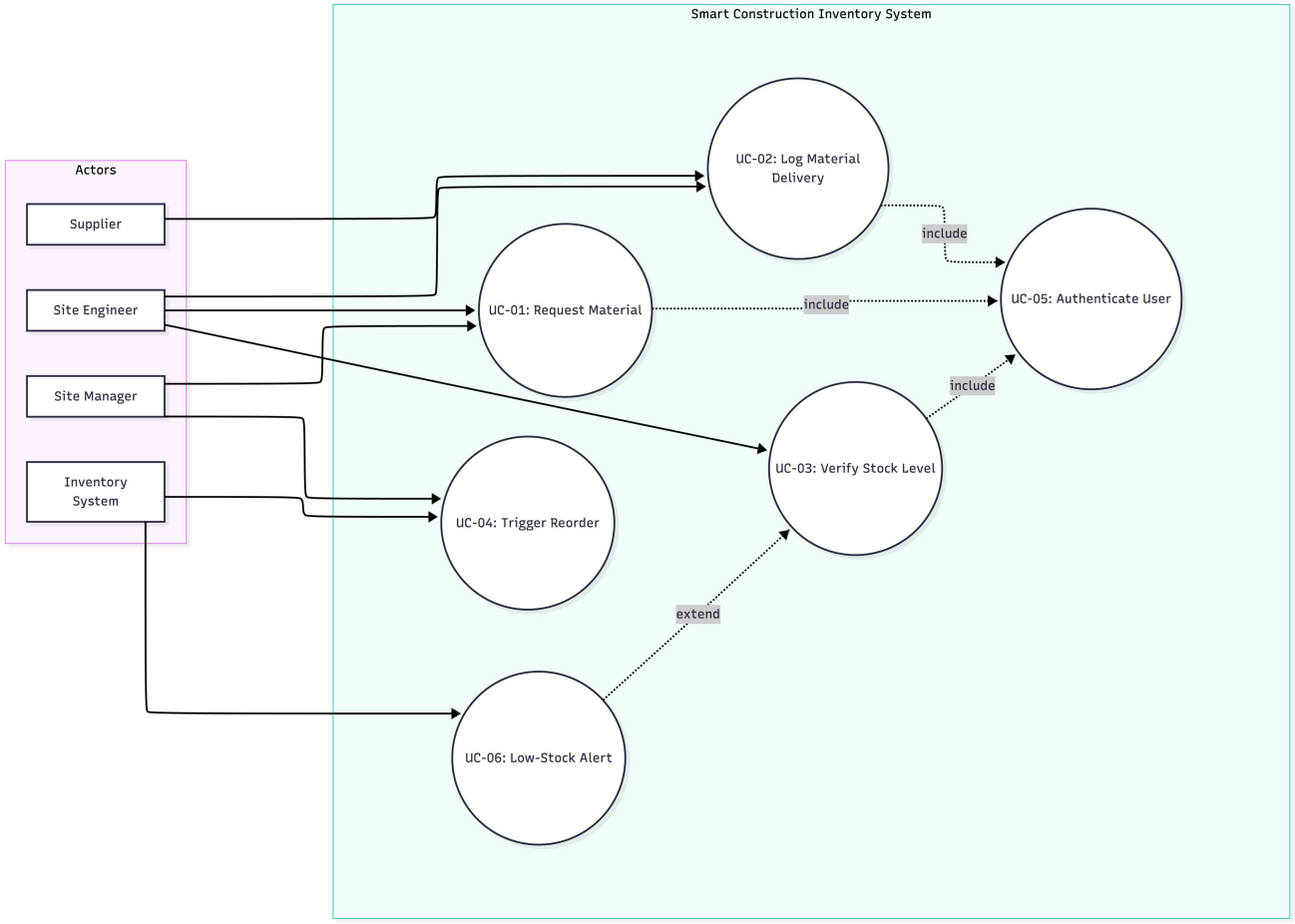


Figure 1: *Use Case Diagram & System Boundary Map (Actors, Includes, and Extensions)*

2. Business Goals & Non-Functional Requirements (NFRs)

The primary purpose of the Smart Construction Site Material & Inventory Management System is to transform manual, error-prone site paper logs into a real-time, auditable digital workflow. Grounded in software engineering practices, clear business objectives and non-functional requirements (NFRs) serve as measurable baselines for architectural decisions and system validation.

2.1 High-Level Business Objectives

Goal ID	Business Objective	Target Impact & Metric
BG-01	Material Wastage Reduction	Reduce raw site material wastage, spoilage, and theft by 20% within 6 months.
BG-02	Downtime Prevention	Decrease stockout-induced work stoppages by 35% via automated reorder alerts.
BG-03	Procurement Efficiency	Shorten Purchase Order approval-to-dispatch lead time from 48 hrs to < 4 hrs.
BG-04	Inventory Reconciliation	Achieve 98% accuracy between physical audit counts and digital inventory logs.

2.2 System Non-Functional Requirements (NFRs)

NFR ID	Quality Attribute	Specification & Verification Criteria
NFR-01	Availability & Uptime	Backend APIs shall maintain 99.9% uptime during operational site hours (06:00 to 22:00).
NFR-02	Performance & Latency	API query response times for stock balance checks shall be ≤ 2.0s under 500 concurrent users.
NFR-03	Offline Synchronization	Mobile local DB (SQLite) shall store up to 1,000 offline transactions & sync within 30s of reconnection.
NFR-04	Security & Data Integrity	Data in transit encrypted via TLS 1.3; local store uses AES-256 with Role-Based Access Control (RBAC).
NFR-05	Usability & Field UI	High-contrast mobile interface with large touch targets (≥ 48px) optimized for outdoor glove usage.

3. Software Architecture Definition & Data Flow

To support field mobile devices operating in remote construction yards with intermittent cellular connectivity, the system adopts a 3-Tier Layered Client-Server Architecture. This establishes a clean separation of concerns between user interaction, business rule processing, and data persistence.

3.1 Architectural Flowchart & System Layers

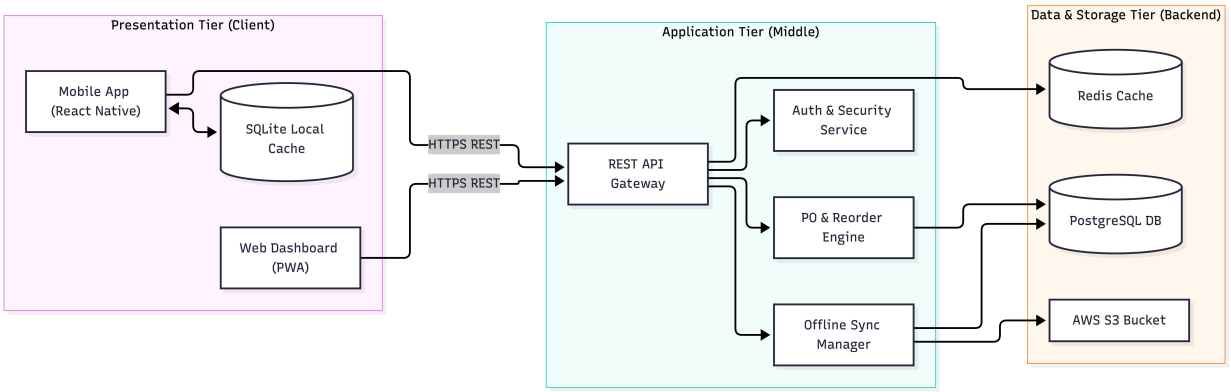


***Figure 2**: 3-Tier Layered Architecture & Data Flowchart (Client Tier, Middle Tier, Backend Tier)*

Layer Tier	Technology Stack	Core Responsibilities
Presentation Layer (Client Tier)	Progressive Web App (PWA), React Native, SQLite	Mobile QR docket scanning, local transaction caching, high-contrast field UI rendering.
Application Layer (Middle Tier)	Node.js REST API, Express, Offline Sync Engine	Executes business logic, authenticates JWTs, processes low-stock reorder velocity models.
Persistence Layer (Backend Tier)	PostgreSQL, Redis Cache, AWS S3 Object Store	Persists relational domain entities, provides sub-second stock caching, stores weighbridge photo proofs.

4. Use Case Analysis & Prioritization

System functionality is partitioned across primary human actors (Site Engineer, Site Manager, Supplier) and automated system actors. To maximize early business value, use cases are prioritized using the MoSCoW framework.

4.1 MoSCoW Prioritization & Release Roadmap

MoSCoW Category	Use Case Name	Business Rationale	Release Target
Must Have	UC-02: Log Material Delivery UC-03: Verify Stock Level	Core field operational need; prevents material theft and stock discrepancies.	Sprint 1 (MVP)
Must Have	UC-05: Authenticate User UC-04: Trigger Reorder	Essential security baseline and critical automation to prevent work stoppages.	Sprint 1 (MVP)
Should Have	UC-01: Request Material UC-06: Low-Stock Alert	Automates internal approval workflows and alerts engineers proactively.	Sprint 2
Could Have	Supplier Performance Rating	Analytically grades suppliers based on lead time accuracy and defect rates.	Sprint 3

5. Documented Significant Use Cases (Cockburn Format)

5.1 Use Case Specification: UC-02 – Log Material Delivery

Use Case ID & Name	UC-02: Log Material Delivery
Primary Actor	Site Engineer / Gatekeeper
Preconditions	1. Engineer logged in via UC-05. 2. Purchase Order exists in 'Dispatched' state in DB (or cached locally).
Main Success Scenario (Happy Path)	1. Engineer selects 'Receive Delivery' on mobile app. 2. Engineer scans QR code on supplier's delivery docket or enters PO ID manually. 3. System displays PO details (Expected Material, Ordered Qty, Supplier Name). 4. Engineer inspects shipment and enters Delivered Qty, Batch No., and Storage Yard. 5. Engineer attaches photo of weighbridge slip or quality test docket. 6. System validates quantity, creates DeliveryReceipt record, updates InventoryLog balance, and marks PO status as 'Delivered'.
Alternate Flows	4a. Partial Delivery: Engineer marks 'Partial Delivery'; system updates stock balance and sets PO status to 'Partially Delivered'. 5a. Quality Defect Found: Engineer flags 'Rejection' with photo proof; system generates Rejection Slip and blocks stock addition.
Exception Flows	3a. Offline Network Mode: App detects zero cellular signal and saves delivery record to encrypted local mobile DB (SQLite). System auto-syncs record to cloud backend upon network restoration (NFR-03).
Postconditions	1. Material stock balance incremented in database. 2. DeliveryReceipt record generated and linked to PO ID.

	3. Audit log updated with timestamp and GPS coordinates.
Business Rules & NFRs	BR-01: Delivered quantity cannot exceed 105% of PO quantity without Manager approval. BR-02: Weighbridge photo mandatory for aggregate deliveries > 5 metric tons.

5.2 Use Case Specification: UC-04 – Trigger Reorder & Process Purchase Order

Use Case ID & Name	UC-04: Trigger Reorder & Process Purchase Order
Primary Actor	Site Manager / Inventory System (Automated)
Preconditions	1. Safety stock limits (Min Threshold, Reorder Qty) configured per material. 2. Approved Supplier catalog linked to material entity.
Main Success Scenario (Happy Path)	1. Inventory System detects stock level ≤ minimum safety threshold during stock verification. 2. System triggers UC-06 Low-Stock Alert and generates a 'Draft Purchase Order'. 3. System transmits alert notification to Site Manager's dashboard. 4. Site Manager reviews Draft PO, verifies budget, and clicks 'Approve PO'. 5. System updates PO status to 'Approved' and transmits digital PO payload to Supplier. 6. Supplier system acknowledges PO and updates status to 'Dispatched'.
Alternate Flows	4a. Manager Adjusts Quantity: Manager modifies order quantity based on pouring schedule before approval. 4b. Manual Override: Engineer triggers manual reorder due to impending weather delay.
Exception Flows	5a. Supplier Out of Stock: Supplier API returns 'Capacity Exceeded'; system alerts Manager and suggests secondary contracted Supplier from master database.
Postconditions	1. New Purchase Order created in 'Approved' or 'Dispatched' state. 2. Audit log entry recorded with manager approval stamp.
Business Rules & NFRs	BR-03: Orders > $50,000 require dual approval (Site Manager + Project Director). BR-04: Auto-reorder suppressed if storage yard capacity > 90% full.

6. Behavioral UML Modeling & Interactive Workflows

Behavioral UML models depict dynamic system behavior over time, capturing workflow decisions, component state transitions, and interaction sequences between objects.

6.1 End-to-End Procurement & Delivery Activity Flowchart

The activity diagram models the end-to-end workflow partitioned into three distinct swimlanes: **Site Engineer | Inventory System | Supplier**.

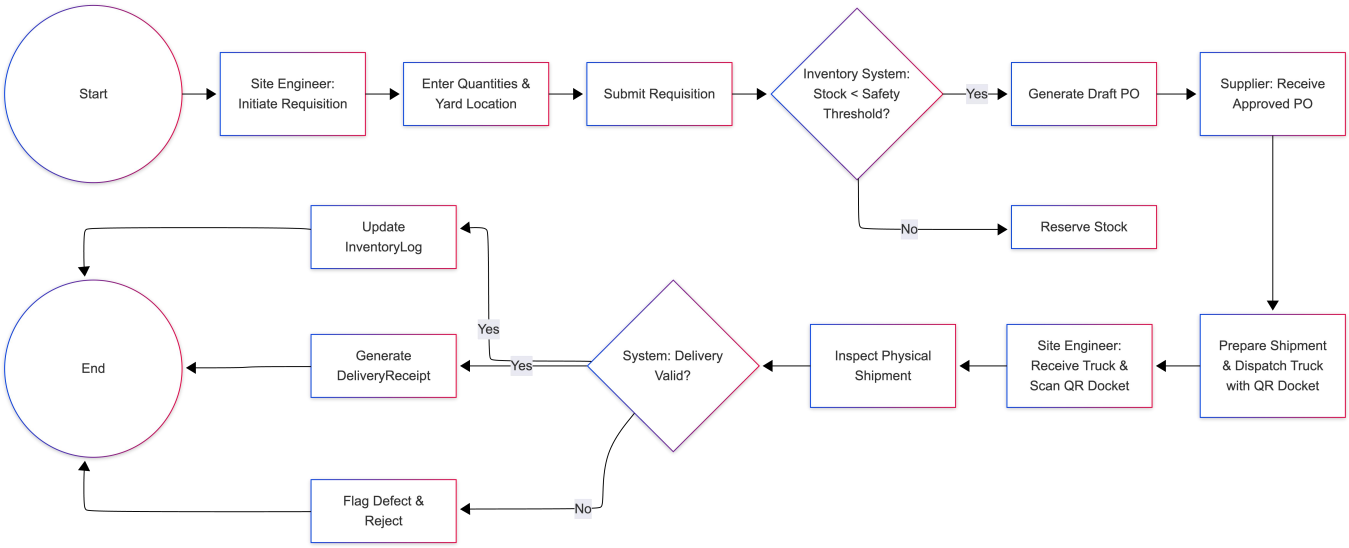


Figure 3: *Multi-Swimlane Activity Diagram (Site Engineer | Inventory System | Supplier)*

6.2 Purchase Order Lifecycle State Machine Flowchart

The Purchase Order entity undergoes well-defined state transitions during its lifecycle driven by specific system triggers and guard conditions:

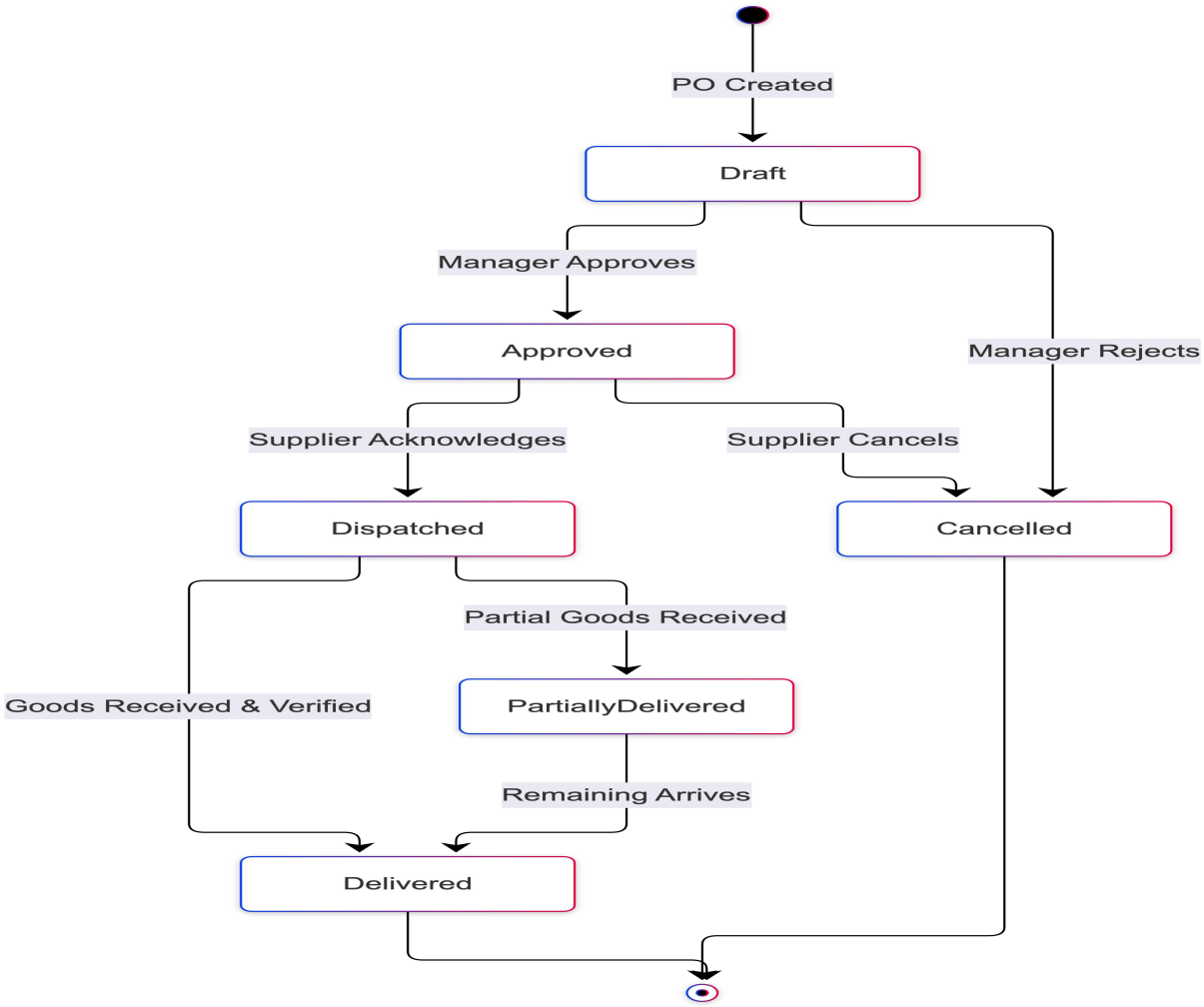


Figure 4: *State Transition Diagram for Purchase Order Lifecycle (Triggers, Guards, & States)*

6.3 Material Delivery Verification Interaction Sequence Flowchart

The sequence diagram specifies time-ordered interaction messages between objects during material delivery verification:

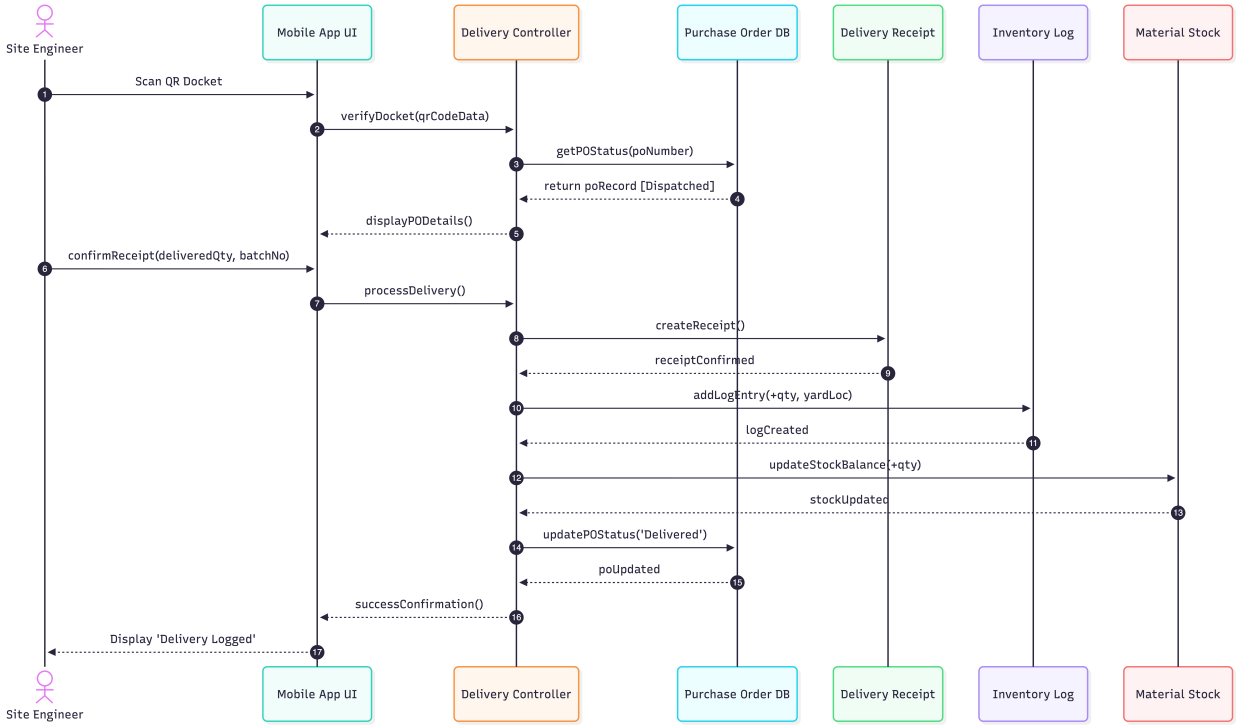


Figure 5: *Time-Ordered Interaction Sequence Diagram for Material Delivery Logging*

7. Domain Class Modeling & Static Structure

Domain class modeling represents the static object structure, specifying domain entities, attributes, operations, and relationships (Composition, Aggregation, Association) with explicit multiplicities.

7.1 Noun Extraction & Candidate Class Disposition

Candidate Noun	Disposition	Design Rationale
SiteManager	Domain Class ✓	Primary human role managing site operations, PO approvals, and threshold overrides.
Supplier	Domain Class ✓	External entity supplying raw materials; tracks contact and rating.
Material	Domain Class ✓	Core inventory entity (cement, steel, aggregate) with safety stock parameters.
PurchaseOrder	Domain Class ✓	Transaction record representing material orders, totals, and lifecycle statuses.
InventoryLog	Domain Class ✓	Auditable transaction ledger tracking stock movements (IN/OUT) and yard locations.
DeliveryReceipt	Domain Class ✓	Verification docket created during site delivery logging with photo attachments.

7.2 UML Domain Class Model Diagram

The system models site **management, suppliers, materials, purchase orders, inventory transactions, and delivery receipts** with their key attributes and operations.

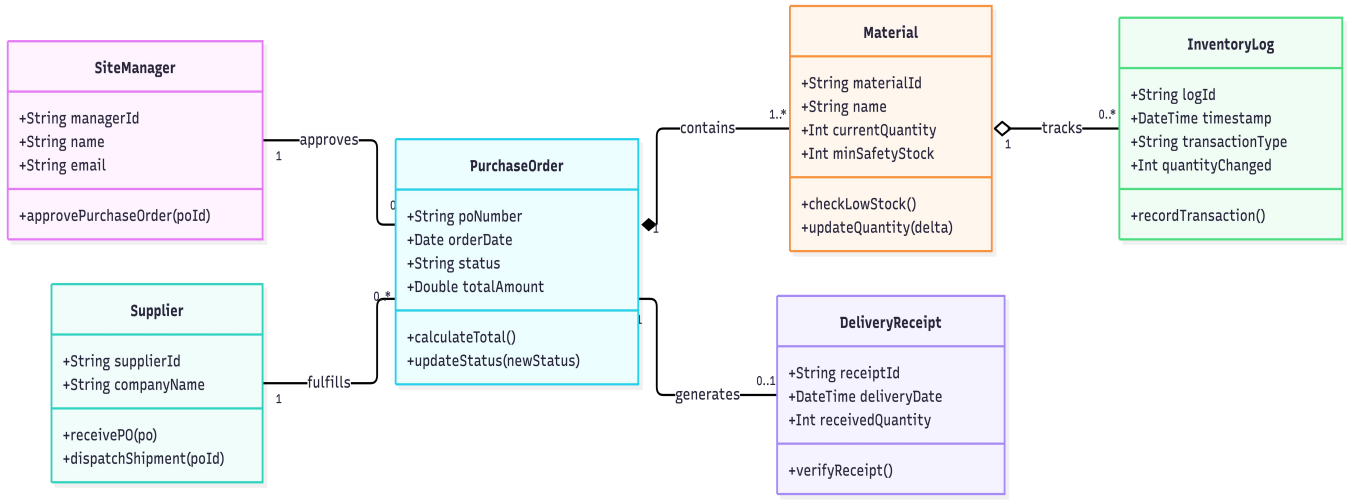


Figure 6: *Domain Class Diagram (Classes, Attributes, Operations, & Multiplicities)*

8. Requirements Traceability Matrix & Concluding Remarks

To ensure complete software engineering rigor, the Requirements Traceability Matrix (RTM) maps business goals to use cases, domain classes, architectural layers, and test cases.

Business Goal	Use Case	Domain Class	Arch. Layer	Test Case ID
BG-01 (Wastage)	UC-02 Log Delivery	DeliveryReceipt, Log	Presentation (Mobile)	TC-DEL-01 (QR Log)
BG-02 (Downtime)	UC-04 Trigger Reorder	Material, Order	Application (Sync)	TC-ORD-02 (Auto Reorder)
BG-03 (PO Speed)	UC-01 Request Material	SiteManager, Supplier	Application (REST API)	TC-REQ-03 (PO Workflow)
BG-04 (Accuracy)	UC-03 Verify Stock	InventoryLog, Material	Persistence (PostgreSQL)	TC-INV-04 (Offline Sync)

CONCLUDING SUMMARY & SUBMISSION READINESS

This enhanced assignment specification provides an interactive, visual-first software engineering blueprint for the Smart Construction Site Material & Inventory Management System. By combining Page 1 Executive Performance Cards, embedded high-resolution UML flowcharts (Use Case, 3-Tier Architecture, Swimlane Activity, State Machine, Sequence Flow, and Class Diagram), and complete Cockburn use case templates, the deliverable provides a testable foundation for system deployment.